

MIGUEL ROSATO CEng · FEI

Completions & Workover Engineering Lead · HPHT & Deepwater Subsea Technical Authority

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Italian citizen (EU) · English · Italian · Spanish

PROFESSIONAL SUMMARY

Completions & Workover Engineering Lead with 25 years designing and delivering HPHT and deepwater subsea completions across the North Sea, Bay of Bengal, Gulf of Suez and the Mesopotamian Basin. Technical authority on DST campaigns up to 14,500 psi / 380°F, sand-control lower completions in 2,000 m water depth, and sour HPHT development wells with H₂S up to 40%. UK Chartered Petroleum Engineer (CEng), Fellow of the Energy Institute (FEI), IWCF Level 4 Supervisor (Combined Surface/Subsea). Currently leading Completions & Workover delivery at BECL Rumaila (1.4 MMbbl/d) across 16 turn-key drilling rigs and 13 workover rigs. Upgrading technical authority with a Professional Certificate in Data Analytics (Imperial College London, May 2026) to drive predictive maintenance and optimisation programmes in wells operations.

CORE COMPETENCIES

HPHT Completions & DST Design · Deepwater Subsea Completions · Sand Control: OHGP, CHGP, Alternate-Path Gravel Pack · Tubular Stress Analysis · Material Selection & OCTG Procurement · Workover Engineering & Well Intervention · Rig-Fleet Management & Scheduling · ESP Design & Diagnostics · Well Integrity · Production Optimisation · Risk Assessment & Well-on-Paper Facilitation · Contract Strategy & Tender Evaluation · M&A Technical Due Diligence · Data Analytics (Python, ML) · Budget Management (\$10M–\$135M)

SELECTED CAREER ACHIEVEMENTS

- **Rumaila Field (2024–2025):** scaled workover deliveries from 180 to 250 per year with the same rig count (+40% throughput); delivered +142,000 bbl/d incremental oil recovery; unlocked ~US\$100M CAPEX/OPEX savings via 13Cr OCTG reutilisation programme; cut rig-move time from 3 days to 1 day.
- **KG-D6 Deepwater HPHT (BP / Reliance, Bay of Bengal):** delivered R-Cluster, SAT-Cluster and MJ field developments — 24 deepwater wells in up to 2,000 m water depth, 10 ksi reservoirs, with alternate-path open-hole gravel-pack lower completions and 7" × 5½" upper completions.
- **Culzean HPHT DST (Maersk Oil, North Sea UK):** prepared and supervised DSTs at 14,500 psi reservoir pressure and 380°F (193°C); managed a £30M+ programme end-to-end from AFE and basis of design through procurement, risk assessment and offshore execution.
- **Pan-Malaysia Deepwater Wellhead Standardisation (Petronas, 2012–2019):** technical lead for a standardised deepwater subsea wellhead and casing-schedule contract adopted across five operators (Petronas, Shell, Murphy Oil, JX Nippon, Total) under the Deepwater Technical Committee; managed Completions portfolio contracts totalling ~US\$135M.
- **Published HPHT authority:** six peer-reviewed SPE papers as co-author on HPHT completion design, extreme-underbalanced perforating, high-gas-rate carbonate wells and 17Cr CRA OCTG first deployment; SPE instructor for the short course "Introduction to HPHT Completions", Kuala Lumpur, 2018.

PROFESSIONAL EXPERIENCE

Completions & Workover Engineering Lead

Jan 2024 – Present

Basra Energy Company Limited (BECL) — BP / CNPC Joint Venture — Basra, Iraq

- Lead Completions & Workover engineering for the 1.4 MMbbl/d Rumaila Field across 16 turn-key drilling rigs and 13 workover rigs.
- Manage a 40-person engineering function: 6 expatriate Senior Workover/Stimulation Engineers, 2 expatriate Senior ESP Specialists, 6 local Senior Workover Engineers and 26 local Workover Engineers.
- Delivered +40% workover throughput (180 → 250 per year) with the same rig count and +142,000 bbl/d incremental oil recovery between 2024 and 2025.
- Introduced 13Cr OCTG reutilisation programme, generating ~US\$100M CAPEX/OPEX savings; redesigned field management of the workover fleet to cut rig-move time from 3 days to 1 day.
- Interface with Subsurface on compartmentalised water injection, water shut-off, well recalibration, big-bore redesign (9½" production casing), perforation optimisation and DST design for sour HPHT reservoirs (>10 ksi, high H₂S).
- Oversee well-integrity monitoring across ~1,600 wells, with coiled tubing, slickline, braided-line and wireline-logging programmes.

Senior Lead Completions Engineer

Jan 2019 – Jan 2024

BP (seconded to KG-D6 JV with Reliance Industries Limited) — Navi Mumbai, India

- Led deepwater HPHT appraisal and completions design for the KG-D6 block (Bay of Bengal, water depth up to 2,000 m, reservoir PVT within the HPHT envelope).
- Managed a team of 4 Senior and 4 Junior Completion Engineers focused on alternate-path open-hole gravel-pack lower completions, upper completions, well testing and well services (slickline, wireline, coiled-tubing stimulation).
- Delivered R-Cluster and SAT-Cluster developments: 12 deepwater wells, ~2,000 m WD average, 10 ksi, OHGP lower completions with 7" × 5½" uppers.
- Delivered a further 12 wells on the MJ HPHT deepwater development.
- Accountable for workover execution across producing deepwater fields.

Subject Matter Expert — Completions

Aug 2012 – Jan 2019

Petronas Carigali Sdn. Bhd. — Kuala Lumpur, Malaysia

- Technical Authority for Completions, Workover and Wellheads across Petronas' offshore portfolio: engineering peer review, risk assessment, well-on-paper facilitation, detailed completion programmes, on-site supervision of HPHT and deepwater operations.
- SME lead across three concurrent multidisciplinary project teams covering deepwater (1,000–3,000 m WD), HPHT development and Ultra-HPHT DST design.
- Technical lead for a dedicated Subsea Deepwater Wellhead contract with standardised casing schedule adopted across five operators (Petronas, Shell, Murphy Oil, JX Nippon, Total) under the Pan-Malaysia Deepwater Technical Committee.
- Authored Petronas Technical Manuals for Completion Design, HPHT, Wellhead / Xmas Tree Design and Subsea Completion Design; coached Drilling and Petroleum Technology engineers in completion, workover and intervention design.
- Submitted, awarded and managed contracts totalling ~US\$135M (2012–2016).

Senior HPHT Completion Engineer (Consultant)

May 2011 – Aug 2012

BP — Sunbury, United Kingdom

- Delivered HPHT DST design for appraisal wells and HPHT completion design for sour development wells (H₂S up to 40%).
- Managed Completion, TCP, Well Testing and Coiled Tubing contracts totalling ~US\$34M.
- Conducted service-company audits of CNPC units in China (Jiangsu wellheads; Bohai coiled tubing / stimulation / well testing; GWDC coiled tubing / TCP / well testing; Daqing coiled tubing / stimulation), covering HSE, QMS, base management, experience, equipment specification and personnel capability.

Senior Completions & Well Testing Engineer

Apr 2010 – May 2011

Maersk Oil North Sea UK Ltd. — Aberdeen, United Kingdom

- Prepared and supervised the Culzean HPHT DST campaign — reservoir pressure 14,500 psi, temperature 380°F (193°C).
- Ran the programme end-to-end: AFE, basis of design, equipment selection and procurement, DST programme, risk assessment and mitigation, budget control in excess of £30M.

Senior Subsea & HPHT Completions Engineer

Nov 2008 – Apr 2010

Total E&P UK — Aberdeen, United Kingdom

- Led budget, programme preparation, material procurement and offshore operations supervision for a subsea workover with dual-ESP systems.
- Led subsea light well-intervention programmes end-to-end.
- Supervisory delivery: Nuggets subsea highly-deviated OHGP completion; Jura HPHT subsea workover; Kessog HPHT subsea completion and well test; Elgin G11 HPHT platform well completion.

Completions Engineer

Jun 2006 – Nov 2008

Eni Group — PETROBEL (Belayim Petroleum, Eni subsidiary in Egypt) — Cairo, Egypt

- Designed completion programmes for OHGP, CHGP, expandable sand screen and conventional completions across East Nile Delta offshore concessions.
- Wrote 15-ksi HPHT well-test programmes for exploration wells; designed subsea intelligent completions; prepared workover programmes; oversaw slickline, wireline, coiled-tubing and fishing operations.
- Served as completion/workover superintendent offshore; led well-performance analysis during clean-up to estimate formation damage.
- Managed budgets in excess of US\$125M for completion, well testing and well intervention; led tender technical and commercial analysis for drilling contracts.

Drilling & Completion Engineer

Jun 2005 – Jun 2006

Eni Group — Nigerian Agip Oil Company (NAOC) — Port Harcourt, Nigeria

- Completion and workover programme preparation and execution supervision for the Nigerian Gas Master Plan on OML 60 / 61 / 62 (ENI operator).

Drilling & Completion Engineer

Oct 2004 – Jun 2005

ENI S.p.A. — Exploration & Production Division — San Donato Milanese, Italy

- Well casing and completion design, equipment and material selection, bid preparation; multiphase pressure-loss and flow analysis on vertical tubing.
- Gas-lift side-pocket-mandrel depth estimation (Dacion Field, Venezuela); ESP setting-depth estimation and artificial-lift well-performance analysis (Belayim Field, Egypt; Dacion Field, Venezuela).

MSc Petroleum Engineering (full-time, with Eni E&P placement)

Aug 2003 – Oct 2004

Politecnico di Torino — Turin, Italy

- Placement at Eni E&P: feasibility study of plastic-coated 5½" gas-injector tubulars for friction-loss reduction, Kashagan Field (Kazakhstan); selection of permanent downhole P/T monitoring system for Kashagan.

Wireline Field Engineer

Apr 2002 – Aug 2003

Schlumberger Oilfield Services — Indonesia, UAE & Oman

- Delivered open-hole and cased-hole logging operations across three countries; interpretation and networked pipeline data handling.

EDUCATION

Professional Certificate in Data Analytics — Imperial College London, London, United Kingdom 2025 – May 2026

Master in Business Administration (MBA) — IE Business School, Madrid, Spain 2008 – 2009

MSc Petroleum Engineering — Politecnico di Torino, Turin, Italy 2003 – 2004

B.Eng. Mechanical Engineering — Honours — Universidad Central de Venezuela, Caracas, Venezuela 1995 – 2001

CHARTERED STATUS, CERTIFICATIONS & PROFESSIONAL AFFILIATIONS

- **Chartered Engineer (CEng)** — **Chartered Petroleum Engineer**, UK Engineering Council — Reg. No. 608108 — since 2013.
- **Fellow of the Energy Institute (FEI)** — Reg. No. 45637 — since 2013.
- **IWCF Well Control Level 4, Supervisor** — Combined BOP Stack Surface / Subsea — valid since February 2026.
- **SPE Member No. 3157452** — since 2004.
- Offshore & HSE: Firefighting; H₂S Escape & Survival; Helicopter Underwater Escape Training (HUET); Offshore Survival; Basic First Aid.

PUBLICATIONS & TECHNICAL INSTRUCTION

- **SPE Instructor** — short course "Introduction to HPHT Completions", Kuala Lumpur, August 2018.
- **SPE-186236-MS** — Cutting-Edge Completion Design for KNUUD: Pioneering HPHT Field Development in Malaysia (co-author).
- **SPE-186948-MS** — Extreme Underbalanced HPHT Coiled-Tubing-Conveyed Perforating: KN-Ultra Deep Field Study (co-author).
- **SPE-186940-MS** — Innovative Completion Strategy Overcomes Operational Challenges in Total-Losses Conditions, HPHT Field (co-author).
- **SPE-191087-MS** — Innovative Solution for High-Temperature High-Gas-Rate Carbonate Well (co-author).
- **SPE-28557-MS** — Pushing Operational Limits in Challenging Perforating Environments (co-author).
- **SPE-28317-MS** — First Successful Deployment of 17-Chrome OCTG CRA Production String (co-author).

TECHNICAL SOFTWARE

Well engineering: Landmark Engineer's Desktop Suite (Compass, WellCat, StressCheck, WellPlan); Schlumberger Wellbarker; Petroleum Experts Prosper; Kappa Saphir; Packpro (sand control); ERA (torque & drag). **Data analytics:** Python (pandas, NumPy, scikit-learn, XGBoost, LightGBM, PuLP, lifelines, matplotlib, seaborn); SQL via Python (SQLite / PostgreSQL). **Project & document management:** Oracle Primavera P6; Peloton Wellview; Autodesk AutoCAD & Inventor; Microsoft 365.

LANGUAGES & GLOBAL EXPERIENCE

Languages: English (advanced) · Italian (mother tongue) · Spanish (mother tongue). **Citizenship:** Italian (European Union).

Operational experience in 12 countries: Venezuela, Indonesia, Egypt, UAE, Oman, Italy, Nigeria, United Kingdom, Iraq, China, Malaysia, India. **Operators:** Schlumberger, Eni, Total, Maersk Oil North Sea UK, BP, Petronas Carigali, Reliance Industries Limited, Basra Energy Company Limited.